

Then (B)–(E) are equivalent and

$$\begin{array}{ccccc}
 ((\text{B}) \text{ and } (\text{G})) & \Longleftrightarrow & (\text{A}) & \xleftrightarrow{\neq} & (\text{B}) & \Longleftrightarrow & (\text{H}) \\
 & & \Downarrow & & \Uparrow \Downarrow & & \\
 & & (\text{F}) & \xleftrightarrow{\neq} & (\text{G}) & &
 \end{array} \tag{1.3}$$

Moreover, there are examples when all of these properties fail.

The following result illustrates the importance of the Vitali–Carathéodory property for some of the properties in Theorem 1.2. It also strengthens Proposition 3.5 in Malý [24], which says that (E) holds under the assumptions in (b).

Theorem 1.5. *Assume that X has the Vitali–Carathéodory property.*

- (a) *If $\mathcal{P}$ is proper (i.e. all closed balls are compact), then $C_X^{\mathcal{P}}$ is an outer capacity, i.e. (B) holds.*
- (b) *If $\mathcal{P}$ is locally compact, then every weakly quasicontinuous $u : \mathcal{P} \rightarrow \overline{\mathbf{R}}$ is quasicontinuous, i.e. (D) holds.*

Theorem 1.1 shows that the Vitali–Carathéodory assumption cannot be dropped even if one merely aimed to show in (a) that $C_X^{\mathcal{P}}$ is an outer capacity for sets of zero capacity (i.e. (E)). We do not know if the assumption that $\mathcal{P}$ is proper can be omitted in Theorem 1.5, nor if the remaining properties from Theorem 1.2 can fail when X has the Vitali–Carathéodory property.

However, we have the following result. That (F) and (G) hold under this assumption was shown by Malý [24, Proposition 4.5]. Note that $\mathcal{P}$ is not assumed to be locally complete here.

Theorem 1.6. *Assume that continuous functions are dense in $N^1X(\mathcal{P})$. Then (A)–(E) are all equivalent and (F)–(G) hold. However, it can happen that (A)–(E) fail.*

Our Example 7.2(a) showing that (A)–(E) can fail even when continuous functions are dense in $N^1X(\mathcal{P})$ is a Banach function lattice (satisfying also (P3) and (P5), but not (P4)). We do not know if this can happen when X is a Banach function space.

Theorems 1.5(a) and 1.6 give the following result, which appeared already in Malý [24, Corollary 4.6 and Propositions 3.5, 4.3, 4.5 and 4.8] (except for (B), which was only deduced under the additional assumption that X is normed).

Corollary 1.7. *If $\mathcal{P}$ is locally compact, X has the Vitali–Carathéodory property and continuous functions are dense in $N^1X(\mathcal{P})$, then all functions in $N^1X(\mathcal{P})$ are quasicontinuous.*

Hence, it follows from Theorem 1.2 that (A)–(G) are all true.

Therefore, at least one of the three assumptions in Corollary 1.7 has to be violated to be able to find a non-quasicontinuous Newtonian function.

Some of the implications in Theorem 1.2 are of course trivial. For the nontrivial ones, Malý [24, Propositions 4.3 and 4.8] has earlier shown that (B) $\Rightarrow$ (D), and for normed X that (A) $\Rightarrow$ (B).

Since the L^p norm, with $1 \leq p < \infty$, has the Vitali–Carathéodory property (see e.g. Malý [24, Proposition 2.1]), Corollary 1.7 recovers several known results for the standard Newtonian $N^{1,p}$ spaces, assuming that $\mathcal{P}$ is locally compact and such that continuous functions are dense in $N^{1,p}(\mathcal{P})$, cf. [5, Section 5.2], [6, Theorem 9.1], [7, p. 1190], [8], [12] and [19, Remark 7.4.3 and Corollary 8.2.5].